Technology Stack

Date	24 June 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

The Technology Stack for this project consists of the programming languages, frameworks, libraries, cloud services, and tools used to develop, deploy, and manage the Credit Card Approval Prediction System. It is designed to provide a secure, scalable, and user-friendly application that predicts credit card approval using machine learning.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML, CSS, Bootstrap, JavaScript	Creates a responsive and user-friendly web interface for entering applicant details and displaying prediction results.

2	**Backend / Server-Side**	Python, Flask	Processes user requests, performs data preprocessing, loads the trained model, and returns approval or rejection predictions.
3	**Database / Data Storage**	CSV Dataset / SQLite	Stores applicant information, historical credit data, and prediction records for analysis and future reference.
4	**Cloud / Hosting / Deployment**	IBM Watson Machine Learning	Deploys the trained machine learning model on the cloud to provide scalable, real-time prediction services.
5	**Version Control & CI/CD**	GitHub	Manages source code, tracks changes, and supports collaboration and version control.
6	**Third-Party APIs / Other Tools**	Scikit-learn, XGBoost, Pandas, NumPy	Scikit-learn and XGBoost are used to train machine learning models, while Pandas and NumPy are used for data preprocessing and analysis.